DAY 32 · THEORY HANDOUT

Find Related Text With Embeddings

Turn text into vectors and build a small FAISS search index.

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Laxmi Kant Tiwari · kgptalkie.com

The idea and its useful terms

Turn text into vectors and build a small FAISS search index.

Embedding

A learned number vector that represents text.

Similarity

A score for how close two vectors are.

FAISS index

A structure that helps find nearby vectors.

Matching exact words can miss a related phrase. Embeddings can place texts with related meanings near each other.

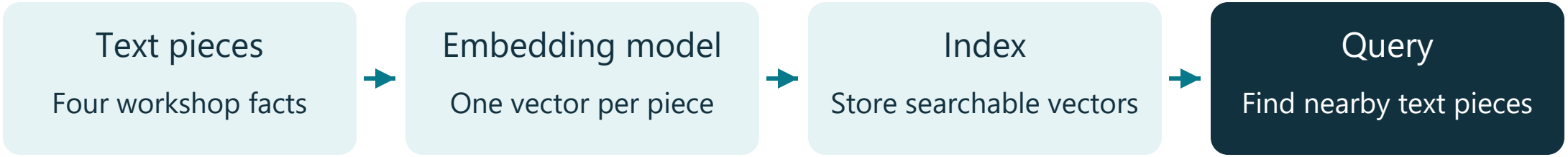
A chat model writes replies. An embedding model makes vectors. We use the same embedding model for documents and questions so their vectors match.

> Nearest means closest among stored items. It does not mean the answer is present or true.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

The embedding model maps each text piece to a fixed-length vector. FAISS searches vectors, while the document store keeps their original words and labels. A returned vector match becomes a readable Document.

The question is embedded with the same model. Mixing models can give different dimensions or incompatible vector spaces. Nearby vectors often suggest related meaning, but a tiny index always has a nearest item.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Saved text	Question	Search result
Pandas works with tables. A fact in the index.	Which tool reads rows? Different wording.	Return the nearby text. Read it before answering.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

> Nearest means closest among stored items. It does not mean the answer is present or true.

Choose the setting or tool for its job

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A different choice changes the work. These are the choices we compare.

Vector

Numbers, not a sentence.

We search with them.

Document text

Words linked to a vector.

We read this result.

Metadata

A source label or page.

We keep the evidence.

Embed four local facts and inspect the vector length.

Build FAISS from the facts and search with a different phrase.

> Try an unrelated question. The nearest item may still be irrelevant.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Try an unrelated question. The nearest item may still be irrelevant.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

> Nearest means closest among stored items. It does not mean the answer is present or true.

DAY 32 · COMPLETE

Thank You

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